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REVIEW

Flood prediction using machine learning and deep learning models: a systematic review

Mohamed Aghenda[1] · Adnane Labbaci[1] · Lhoussaine Bouchaou[1] · Mohammed Hssaisoune[1] · Yassine Ait Brahim[2]

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Abstract
Flood prediction can be challenging due to the complexity and non-linearity of its hazards, making traditional methods limited and less efficient. Thus, the interest in using machine learning and deep learning to overcome these challenges has increased over the years. This systematic review provides an overview of the current state of the flood prediction field using machine learning and deep learning models. It examines its evolution over the past two decades. It provides insights into the most commonly used models for various prediction tasks and timeframes. It also examines the most suitable input data for models adopted by researchers and explores the performance metrics employed. Our study synthesises 237 Scopus-indexed and English-language papers, retaining the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement as the primary reporting framework. The domain has experienced significant growth since 2021, making major contributions in the field of flood prediction. Long Short-Term Memory (LSTM), Artificial Neural Network (ANN), and Convolutional Neural Network (CNN) are the most commonly used models, demonstrating the superior performance of Deep Learning models over Machine Learning models in flood forecasting. Hybrid and ensemble models are found to be very beneficial in terms of prediction accuracy and efficiency, especially when dealing with complex spatiotemporal inputs. Discharge and flow forecasting is the primary prediction task among hydrologists, and rainfall and streamflow stand out as the primary predictor inputs. However, most studies focus on short-term-based prediction, leaving a gap in forecasting long-term floods that can inform large-scale water resources management strategies.

Keywords Flood prediction · Flood forecasting · Machine learning · Deep learning · Systematic review

Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Networks
BPN	Back-Propagation Artificial Neural Networks
CNN	Conventional Neural Network
CRED	Centre for Research on the Epidemiology of Disasters
CSV	Comma-Separated Value
DEM	Digital Elevation Model
DL	Deep Learning
ENSO	El Niño and the Southern Oscillation
EPS	Ensemble Prediction Systems
E-RNN	Entangled Recurrent Neural Network
FCNN	Fully connected Neural Network
FFNN	Feedforward Neural Network
GBM	Gradient Boosting Model
GIS	Geographic Information System
GNN	Graph Neural Network
GRU	Gated Recurrent Unit
HA	Historical Average
IPCC	Intergovernmental Panel on Climate Change
KGE	Kling–Gupta Efficiency
KNN	K-Nearest Neighbours
kNN	K-Nearest Neighbours
LGP	Linear genetic programming
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean absolute percentage error

✉ Mohamed Aghenda
Mohamed.aghenda@edu.uiz.ac.ma

[1] Faculty of Sciences of Agadir, Ibn Zohr University, Agadir, Morocco

[2] Mohammed VI Polytechnic University (UM6P), Ben Guerir, Morocco

ML	Machine Learning
MLP	Multilayer Perceptrons
MSE	Mean Squared Error
NAO	North Atlantic Oscillation
NSE	Nash-Sutcliffe Efficiency
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
QUOROM	Quality of Reporting of Meta-analyses
RF	Random Forest
RIS	Research Information Systems
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
SAINA-LSTM	Self-activated and Internal Attention LSTM
SCS-CN	Soil Conservation Service Curve Number
STA-LSTM	Spatial-temporal attention mechanism LSTM
SVM	Support Vector Machines
SVR	Support Vector Regression
TPR	True Positive Rate
UNDRR	United Nations Office for Disaster Risk Reduction
WNN	Hybrid wavelet neural network
WSN	Wireless Sensor Network
XGBoost	Extreme Gradient Boosting

1 Introduction

The severe impacts of climate change on both natural and human systems, along with the associated risks and vulnerabilities, provide a crucial foundation for understanding the current climate emergency (Fawzy et al. 2020). Significant changes in key climate indicators, including temperature, precipitation patterns, sea level rise, ocean acidification, and extreme weather events, are highlighted in a report by the United Nations Climate Change Secretariat (UNCCS 2019). The report highlights a range of climate hazards that are increasingly affecting ecosystems and human communities worldwide, including droughts (El Bouazzaoui et al. 2024, 2025; Imane et al. 2025), floods, hurricanes, severe storms, heatwaves, wildfires, cold spells, and landslides. Floods are one of the most significant climate-related disasters, causing substantial damage (Dhankhar et al. 2025b) and affecting more people than any other environmental hazard, thereby hindering sustainable development (CRED 2020).

Floods are defined by the Intergovernmental Panel on Climate Change as "the overflowing of the normal confines of a stream or other body of water, or the accumulation of water over areas that are not normally submerged. Floods include river (fluvial) floods, flash floods, urban floods, pluvial floods, sewer floods, coastal floods, and glacial lake outburst floods" (IPCC 2018). According to the IPCC SREX report, flood events have become increasingly severe, prolonged, and frequent due to changes in climate, land use, infrastructure, and demographics (Kundzewicz et al. 2014). Between 2000 and 2019, global flood-related damages were estimated at $651 billion (USD), impacting approximately 1.6 billion people worldwide (2022) (CRED 2022). The projections of climate change, particularly by 2030, suggest that the proportion of the population at risk of flooding will continue to increase (Tellman et al. 2021).

In the context of flood prediction, having warning and alert systems available for the public to take precautions and prepare safety measures is crucial, as it can save many lives and minimise economic losses, especially in urban areas (Xu et al.). Extreme precipitation is considered the most critical factor that causes flooding(Garg and Mishra 2019; Gao et al. 2024), however, flood magnitude and precipitation have a non-linear relationship as the flood phenomenon is considered complex and variable due to the existing interaction between several other triggering factors such as landcover, topography, and evapotranspiration (Gao et al. 2024; Zhang et al. 2025; Ke et al. 2025), these factors generate various types of data, including hydrological data, rainfall trends, infrastructure characteristics and topographical features that suffer incompleteness, inconsistency, and inaccuracy in most cases, thus the need to develop novel methodologies and models integrating Artificial Intelligence (AI) and other advanced technologies (Yu et al. 2025).

In the past, traditional methods for flood prediction were primarily based on physical, statistical, and probabilistic approaches, utilising commonly used hydrological models in conjunction with deterministic and empirical weather prediction models. (Gouweleeuw et al. 2005; Bürger et al. 2009). Other studies have employed various approaches, including the prediction of river stage-discharge using numerical experiments (Abril and Knight 2004), a data mining approach utilising the river's daily discharge time series (Damle and Yalcin 2007), radar-based quantitative precipitation estimates (QPEs) derived from rain gauge data (Morin et al. 2009), and Satellite-based precipitation products to assess the consequences of climate change (Najmi et al. 2023). Archfield et al. (2013) employed an improvement over traditional statistical methods, utilising spatial interpolation techniques that effectively predicted streamflow. Additionally, some studies have employed methods that combine GIS and hydrological models for flood prediction (Mioc 2008).

Other studies used socioeconomic aspects to assess social vulnerability to the risk of flooding. (Bouaakkaz et al. 2023) developed a methodology based on 26 socioeconomic indicators to assess social vulnerability to flooding,

while Ahmadivostakolaei et al. (2024) investigating the social resilience when a natural disaster occurs using structural equation modelling.

Numerical weather predictions (Chen et al. 2019) and ensemble prediction systems (EPS) (Demeritt et al. 2013) were utilised to forecast streamflow and precipitation during flood seasons, aiming to reduce prediction uncertainty. The emergence of advanced techniques was observed using computer programs, such as linear genetic programming (LGP), to predict peak monthly discharges (Tofiq and Guven 2014). Flood conditioning factors are also widely used in flood assessment because they are essential in understanding the nature of the process and its dynamics. This approach has been explored in numerous studies that focus on multicriteria decision-making analysis, often in conjunction with GIS techniques, ensemble methods, or Machine Learning (ML) models (Khosravi et al. 2019; Ali et al. 2020; Gudiyangada Nachappa et al. 2020).

Recent innovations in remote sensing technologies and ML models offer a transformative approach to flood detection and assessment challenges. Remote sensing provides a cost-effective and extensive means to capture real-time and historical data on land use, water levels, and surface changes over large, inaccessible areas. Using satellite imagery, aerial photography, and radar data, researchers can track flood events with high temporal and spatial resolution. Combined with ML algorithms and various datasets, these technologies enable accurate, real-time flood forecasting, which is essential for mitigating damage to human property and planning effective defences (Zhang et al. 2023).

There are existing studies that have conducted surveys on the applications of ML in flood prediction. Mosavi et al. (2018), for instance, a state-of-the-art review of the application of ML models in flood prediction was provided, focusing on lead-time prediction at specific sites, but excluding the spatial flood prediction aspect. Maspo (2020) systematically examined existing research through the evaluation of input parameters used for flood prediction, and Refadah et al. (2025) investigated the use of ML models in different flood forecasting scenarios, including susceptibility and risk assessment. Our main goal is to produce an updated systematic review of existing studies that used ML and DL models for flood prediction, covering all aspects, including research trends, the most frequently used models, both lead-time and spatial prediction, and the input data used in a quantitative and qualitative structure. This aims to provide a clear insight into the most suitable ML/DL models, input data, and approaches that researchers and hydrologists may follow to address different flood prediction challenges.

Table 1 The search query

Search query
TITLE-ABS-KEY (flood* AND (predict* OR forecast*) AND ("machine learning" OR "deep learning")) AND PUBYEAR > 2004 AND PUBYEAR < 2026 AND (LIMIT-TO (PUBSTAGE, "final")) AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English"))

2 Methodology

A systematic review is a review that employs explicit, systematic methods to collect and synthesise evidence from studies addressing a well-defined question (Higgins et al. 2019). Moreover, to help conduct a well-structured review, we used the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement as the basis for our methodology. PRISMA, funded by the Canadian Institutes of Health Research in 2009, comprises a 27-item checklist that helps authors enhance the reporting of systematic reviews and meta-analyses (Moher et al. 2009). Moreover, it aims to replace the QUOROM Statement of Reporting of Meta-analyses, which focuses on reporting meta-analyses of randomised controlled trials (Moher et al. 1999).

The PRISMA statement comprises four essential phases: identification of all relevant papers, screening of the data to exclude irrelevant studies, assessment of the remaining records for eligibility, and inclusion of the final studies in the systematic review. The detail of each phase is explained in the following:

To extract all articles relevant to our study on the prediction and assessment of flood risk using ML and deep learning (DL) models, we developed a search string that encompasses the studies forming the basis of our systematic review. We used the online database of "**Scopus**" as our main search engine, as it is the most accurate and the world's largest database offering only high-quality indexed articles, conference proceedings, and books (Baas et al. 2020). The sample comprised classified journal articles in English published between 2005 and 2025, and the search was conducted on August 08, 2025.

To ensure that we do not lose any significant studies during the search process, we attempted to create a search query using direct important keywords: "flood, "predict", "forecast, " "machine learning," and "deep learning." Then, we proceeded to select and screen the non-relevant and relevant articles through the screening process, which will be included in our review (Table 1).

The search results revealed "**2698** articles" that fit the search query criteria. All were exported in "CSV," "RIS," and "BibTex" formats with their citation information, bibliographic information, abstracts, and keywords for further analysis.

Table 2 Research questions and description

Research question	Description	Technique
RQ1: What are the paper's trends in flood prediction using ML and DL models?	Identification of the pattern's publication, the areas of focus, and the directions of research in flood prediction	Quantitative
RQ2: What are the most ML and DL models that are used in flood prediction?	Identification of popular algorithms, used to analyse and predict flood data (single or hybrid…)	Quantitative
RQ3: What is the performance of these models?	Assessment of the performance of ML and DL models. What metrics are used for each ML/DL model	Qualitative
RQ4: What are the limitations and challenges in applying these models to flood prediction?	The issues and challenges that researchers face in implementing ML and DL models for flood prediction, such as data source, computational requirements, and model generalization	Qualitative

Table 3 Inclusion and Exclusion criteria

	Inclusion criteria	Exclusion criteria
Study design	Final and published papers in English between 2005–2025	Case studies, opinion papers, conferences, workshops, and papers in other languages
Study area	Studies conducted in any geographic region or specific, localised areas	Studies conducted in a particular situation (flood that occurs due to non-natural events)
Flood type	Riverine, coastal, or urban flooding	Another type of hazard
Input data	Studies that used geospatial data, hydrological and hydro-meteorological data as their inputs (e.g., rainfall, streamflow, DEM, satellite imagery)	Studies relying solely on statistical, economic, or social data
ML and DL models	Employing at least one ML or DL model	Studies not using any ML or DL models or techniques
Prediction task	Studies focused on temporal flood predictions for a specified horizon (1 h to 24 h, daily, yearly)	Studies that focus solely on flood risk susceptibility/assessment and flood mapping
Performance metric	Report at least one quantitative performance metric	Studies without a quantitative performance metric

A systematic review combines all the empirical evidence that meets predefined eligibility criteria to answer a specific research question (Mancin et al. 2024). Reviewing relevant studies to identify research gaps is essential to address these questions, particularly in ML and DL. The research questions in this study are designed to cover all aspects of our review (Table 2).

The selection of articles for inclusion in a systematic review is a critical process that directly affects the accuracy of the results and their subsequent interpretation. To ensure rigour and transparency in this process, the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram (Stovold et al. 2014), was used to filter the 2,698 articles retrieved from the database search.

For this review, the inclusion and exclusion criteria are described in Table 3:

To ensure transparency with the reader, every systematic review study requires an explicit description of the methods and procedures conducted, so that the same results can be obtained when applying the same methodology and reaching the same conclusions (Meline 2006). The process began with identifying relevant articles through a comprehensive search using the search query (Fig. 1) in the advanced search of the Scopus Database Website.

Duplicates were then identified and excluded; only one duplicate was found among the papers, leaving 2697 records retrieved. A title-based screening was then conducted to remove articles that explicitly stated in the title that they did not involve flood prediction or flood forecasting, and had no interest in flood susceptibility and risk assessment mapping or any other unrelated field (Fig. 2).

The abstracts of the remaining articles were then screened to exclude ineligible studies that indicate only susceptibility and hazard mapping, conducting general flood prediction based on hydrology approaches only, and do not indicate that they use at least one ML/DL model for the flood prediction/forecasting task, papers that indicates only the use of non-geographic or hydrological input are excluded in this step. Finally, a full-text review was conducted to analyse the methodology and objectives of each article, to examine the existence of a transparent method with a specific prediction task (Discharge/streamflow, water level prediction…), excluding papers with no clear lead time, or no clear prediction task or that do not use the ML/DL models as their main predictive approach. Papers with no clear evaluation metrics for the used models were also excluded. We finally left with 237 articles for inclusion in the final systematic review.

3 Results

In this section, we will explore the 237 papers included in our systematic review. Publishing trends and Models of ML/DL used in each study will be highlighted, along with the input data used. Prediction tasks, lead times, and performance metrics will be investigated and summarised to address research limitations and gaps, understanding how the field has evolved, and addressing suggestions for future work.

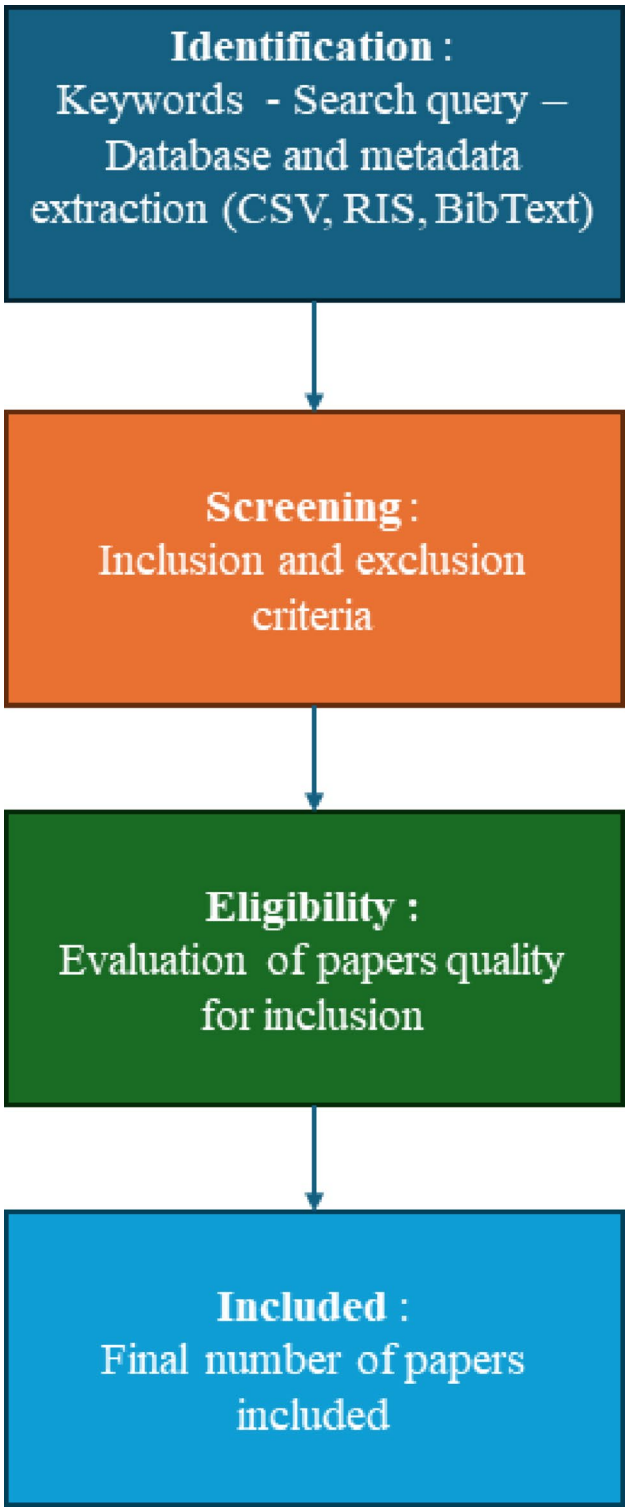


Fig. 1 PRISMA framework

3.1 a. Publication trends, study areas, and keywords:

Figure 3 below shows how research into flood prediction using ML and DL has evolved between 2005 and 2025. Over time, the number of studies has increased significantly. The first of the included papers was published in 2010, as no studies were recorded between 2005 and 2010. This gap can be attributed to the reliance on physical and statistical models for flood prediction research during that period. Between 2010 and 2019, research activity was limited, with only five studies published. From 2020 onwards, the field began to take off, with 15 papers being published in a single year. Following a slight decline in 2021, the number of studies grew rapidly, driven by the increasing adoption of ML and DL across various fields and disciplines. In 2025, there was a slight drop in the number of publications, which can be attributed to the fact that data collection was conducted in the middle of the year (August 8, 2025).

Flood prediction studies have been conducted in various countries worldwide. Figure 4 demonstrates the geographic distribution of the included papers carried out using the Biblioshiny on R project (version 4.5.1). China stands out with the most significant number of papers that have conducted flood prediction studies based on ML/DL models, at a percentage of 38%, followed by India (10%) and the USA (9%). According to the number of studies conducted in each country, the Asian continent dominated, followed by Europe and North America, while Africa, South America, and Australia each contributed about 2%.

Figure 5 illustrates the Word cloud of the keywords from the included papers in the systematic review, visualising the most frequent and mentioned keywords in the dataset. Larger keywords are dominant and more often used in the studies, while smaller ones are less utilised.

Bibliometric mapping using VOSviewer (version 1.6.20) software is essential for visualising the relationships between authors and their keywords in every systematic review (van Eck and Waltman 2010). Figure 6 presents the co-occurrence map, which links terms and keywords from the reviewed papers. Each term is grouped into sets called clusters. Each cluster's size indicates its weight and thus its importance. The links between clusters indicate how an item is related to another item in a different cluster (van Eck and Waltman 2015). We can see in the centre of the map that the terms "flood forecasting", "deep learning" and "machine learning" are most common keywords among authors, which links every other term in the map, among models' term, CNN, LSTM and ANN are the closest to the centre which can indicates the use of these models in most papers. Attention mechanisms, RNNs, LSTMs, and GRUs are strongly linked to streamflow prediction, water level

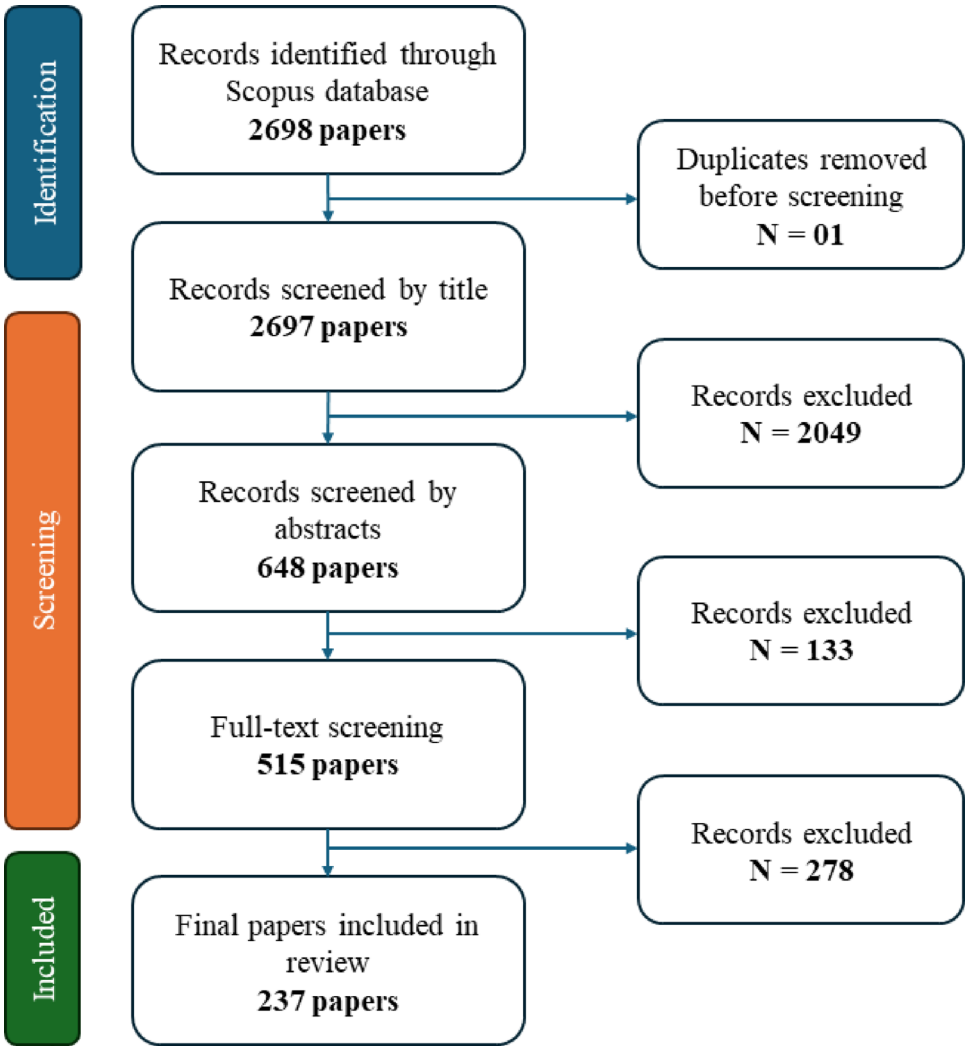


Fig. 2 PRISMA diagram for inclusion and exclusion criteria

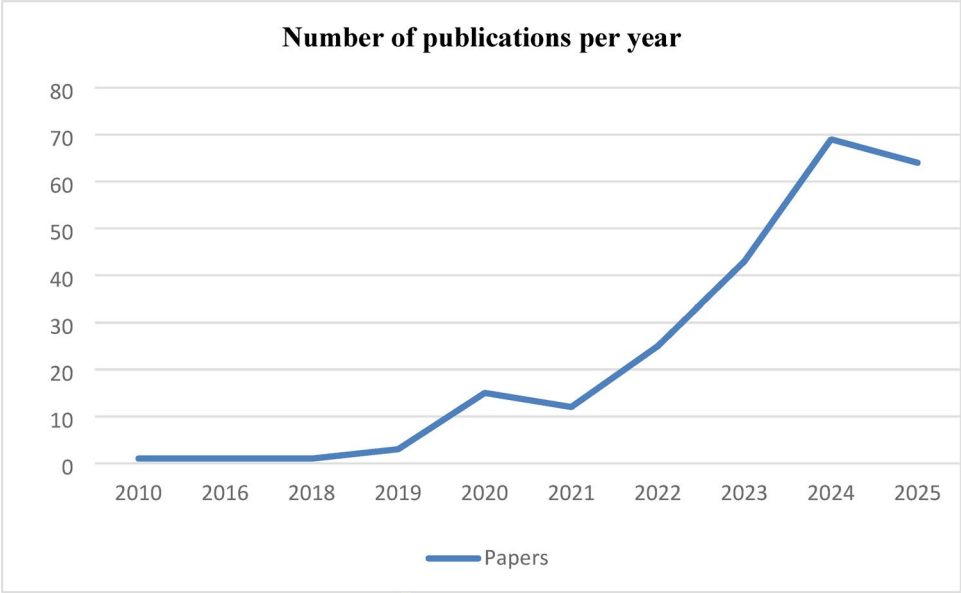


Fig. 3 Number of publications (2010–2025)

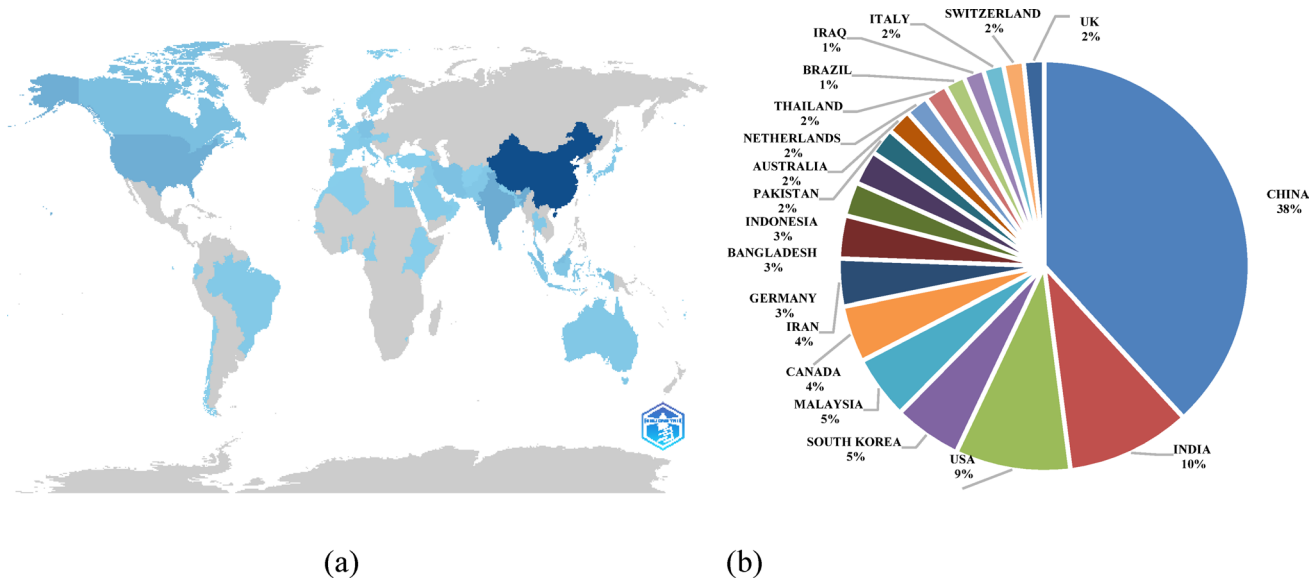


Fig. 4 Distribution of studies **a** Geographical distribution (Biblioshiny), **b** Countries distribution

Fig. 5 Wordcloud of papers keywords (Biblioshiny)

prediction, runoff prediction, hydrological modelling, and real-time flood forecasting, which indicates that DL models are the most widely adopted for forecasting tasks that use time-series data as the primary predictor. At the same time, ANN appears to be closely linked to Water level forecasting. SVM and RF are also visualised as ML models that can be used for flood prediction, but with less significance than most DL models.

To build an ML/DL flood prediction model, researchers often combine historical data of past flood events with real-time data from rain gauge sites. Rainfall and water observations are traditional sources of datasets, utilising ground rain gauges or newer remote sensing technologies, such as satellites, radars, and other multisensory systems. (Mosavi et al. 2018). Lai & Tseng (2010) compared the ability of simple grey models with Back-Propagation Artificial Neural Networks (BPN) and Support Vector Regression (SVR) to forecast one-hour flood using five typhoon events and past water level observations. Considered one of the early studies in the field, it demonstrates that simple grey models can compete with regression models, such as ANN and SVR, in short-lead flood stage forecasting. In 2015, work shifted with the integration of a wireless sensor network (WSN) and Chaos theory, adding continuous rainfall and river data levels from 2013 and 2014 to forecast one-step flood events. This effort compared Multilayer Perceptron (MLP) and Elman Recurrent Neural Network (E-RNN) models (Furquim et al. 2016).

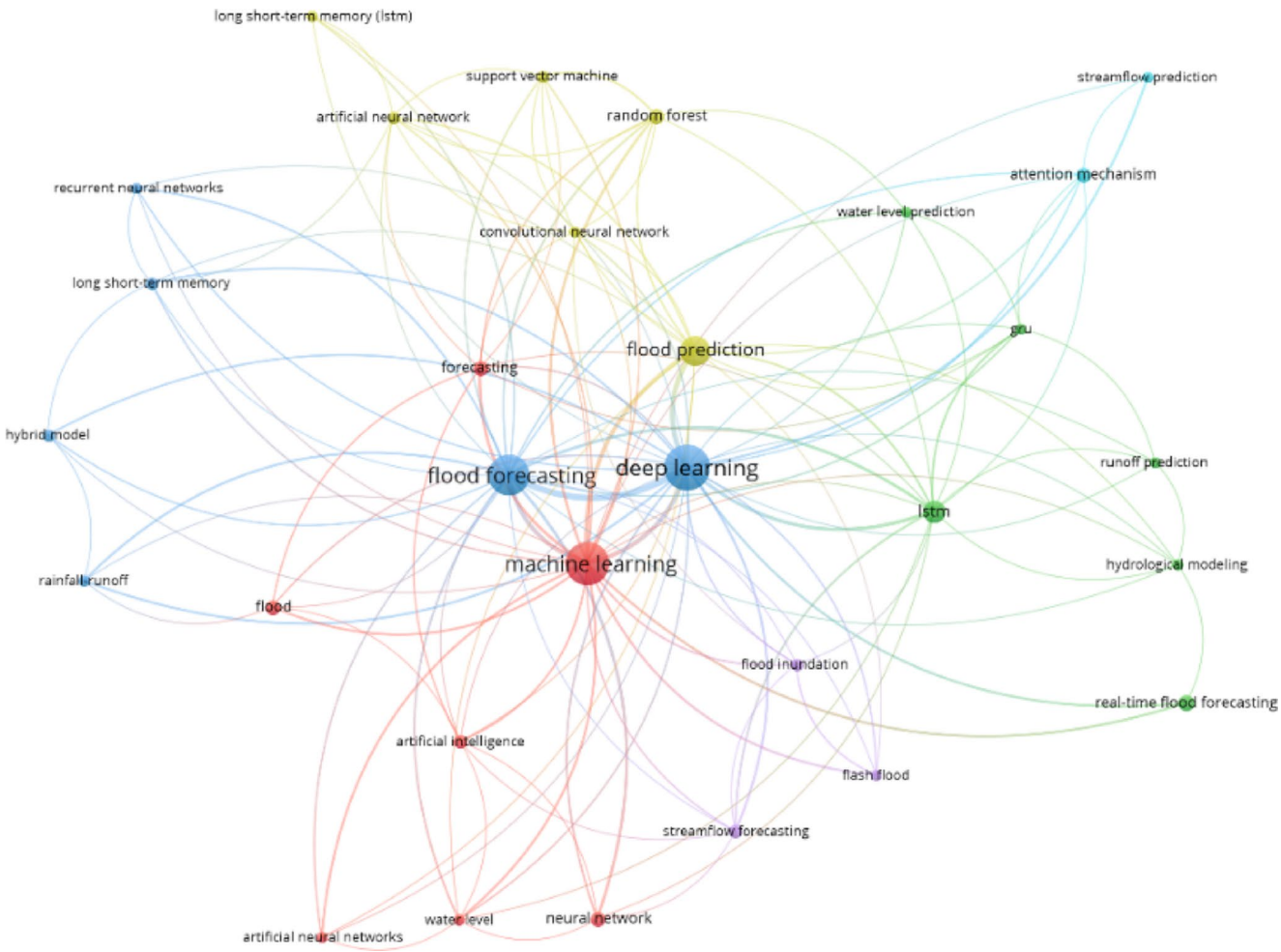


Fig. 6 Co-occurrence map of author keywords with overlay visualisation (VOSviewer)

In flood prediction contexts, using the smallest and simplest dataset other than a large and complex one is always better to make models more robust and can improve predictive skills, this was proved in the work of Muñoz et al. (2018), in which he conducted a Random Forest (RF) modelling process using hourly precipitation from 3 rain gauges along with hourly discharge to forecast 4 to 24 h extreme peak flows. Hybrid models are used to improve accuracy. A hybrid Extreme Learning Machine (ELM) enhanced by K-Nearest Neighbours (KNN) and the Fireworks Algorithm (KNN-FWA-ELM) is employed in (Ren et al. 2019) and demonstrates its effectiveness compared to a plain ELM model. At the same time, researchers are still relying on simple ML models such as XGBoost, RF and SVM (Venkatesan and Mahindrakar 2019). As the decade began, the use of more complex models, such as Long Short-Term Memory (LSTM) and Conventional Neural Network (CNN) (Dng et al. 2020; Song et al. 2020), as well as hybrid models (Anupam and Pani 2020), started to gain attention among researchers. Discharge and flow forecasts remain the main tasks, and the data used still relies on gauge sets as the primary source.

The past overview is important for understanding how the field originated and how ideas evolved, particularly regarding models of ML/DL, data, and sources used to achieve forecast and prediction goals. However, to absorb the trends in the field, the period between 2021 and 2024, with the most publications, is the most important. To clarify these trends, this section will be divided into three sections: Models used, Input data, and Prediction task.

3.2 b. Machine learning and deep learning models:

The most commonly used models are presented in Table 4 along with their methodological basis and applications. Figure 6 shows the use of ML/DL models within the 2010–2025 period in the papers included in our systematic review. Long Short-Term Memory (LSTM) has emerged as the most widely used model for flood prediction tasks over the years, often embedded in hybrid models or augmented with

Table 4 Machine Learning and Deep Learning models

Model	Methodological basis	Applications	References
ANN	Connected neurons with weights in a non-linear form, trained by back-propagation to determine the output signal	Classification/regression and prediction tasks	(Rumelhart et al. 1986)
CNN	Input layer with multiple hidden layers that perform convolutions, generating a feature map that contributes to the input of the next layer, along with pooling layers, fully connected layers, and normalisation layers	Image classification and segmentation, video and image recognition, speech recognition	(Lecun et al. 1998)
LSTM	A recurrent neural network (RNN) that uses a loop structure to process sequential data and retain long-term information via a memory cell. Allowing selective storage and retrieval of information over extended periods	Sequence/time-series forecasting, speech recognition, sensor analytics, natural language processing	(Hochreiter and Schmidhuber 1997)
GRU	A recurrent neural network (RNN) variant with fewer parameters than LSTM that uses update and rest gates to determine how much of the candidate activation vector to incorporate into the new hidden state	Time series forecasting, speech recognition, and natural language processing	(Chung et al. 2014)
RF	A collection of decision trees built based on bootstrap samples with random feature samples	Tabular classification/regression	(Breiman 2001)
SVM	Uses a hyperplane to separate the data from one dimension to high-dimensional space, with non nonlinear transformation in case of nonlinearity of data points	High-dimensional/sparse text classification, image classification	(Cortes and Vapnik 1995)
kNN	Classifies data points based on the metric distance of k nearest points in a dataset	Lightweight classification/regression tasks, recommender systems, anomaly detection	(Cover and Hart 1967)
GBM	Consists of a sequence of weak learners, such as decision trees, by iteratively fitting residuals to minimise loss using gradient descent	High-accuracy tabular prediction, Search ranking, anomaly detection	(Friedman 2001)
XGBoost	Optimised, regularised gradient-boosted decision trees with sparsity-aware blocks and parallelisation	large and structured tabular data predictions, anomaly detection	(Chen and Guestrin 2016)

an attention mechanism to enhance efficiency and focus on the primary inputs in use (Cui et al. 2021; Merizalde et al. 2023; Ebtehaj and Bonakdari 2024; Kordani et al. 2024). One of the most commonly used models is the Convolutional Neural Network (CNN), along with its key components. It is also a challenge to compare LSTM models or use them as a hybrid along with an LSTM to produce better forecasting results (Moishin et al. 2021).

Ding et al. (2020), as the most cited document among the included papers, used an augmented version of the LSTM model, adding a Spatial–temporal attention mechanism (STA-LSTM). This was found to outperform many ML/DL and Hybrid models like the normal version of LSTM and CNN, a fully connected Neural Network (FCNN), and Historical Average (HA). Alizadeh et al. (2021) also used an attention-based Long-Short Term Memory (LSTM) referred to as a Self-activated and Internal Attention LSTM (SAINA-LSTM) and compared it to other ML/DL models and operational forecasting systems. While Moishin (2021) employed a ConvLaSTM hybrid algorithm to forecast future flood occurrences, it outperformed other benchmark models, including LSTM, CNN-LSTM, and SVR.

Artificial Neural Networks (ANNs) are also widely used in the field. They were used by Dtissibe et al. (2020) to forecast discharge and showed good capacity and efficiency for the task, while Linh et al. (2021) found that the hybrid wavelet neural network (WNN) yielded superior results compared to a standalone ANN. Another version of Neural Network tested to predict water levels is the wavelet-based feedforward neural network (FFNN), which, when hybrid, shows superior results compared to singular ones (Kumar and Singh 2024).

RF, as a classic ML model, was one of the first models tested for flood prediction. However, in recent years, much research has found that it compares unfavourably to other singular, ensemble, or hybrid models (Chand et al. 2025;

Cisterna-García et al. 2025; Peng et al. 2025; Soliman et al. 2025). Gated Recurrent Unit (GRU) comes after RF among the tested models. It exhibits moderate performance and should be combined with another Neural network, such as a graph neural network (GNN) or CNNs (Miau and Hung 2020; Alizadeh et al. 2021; Kazadi et al. 2024; Tahermanesh et al. 2025).

Other less-used models are essentially simple ML models, such as SVM, kNN, GBM, and XGBoost, which have become less popular over the years due to the superiority of DL (Sanders et al. 2022; Pungching and Pilailar 2023).

Figure 7 illustrates the challenge between single ML/DL models and hybrid and ensemble models in the context of flood prediction. It shows that DL models in general outperform ML models over the years, with the predominance of single DL models in most case studies.

Since 2019, the use of DL models has become a trend with the emergence of novel models and architectures that handle nonlinear and complex relationships between data (Mienye and Swart 2024). On the other hand, Hybrid DL models are also trending over the years in comparison with single ML and hybrid ML models, and that is due to the advantage that these types of models have, combining multiple strengths of two or more DL models or making an augmented version using techniques like automatic mechanism or hyperparameter optimisation, like grid search (Ng et al. 2023; Hasan et al. 2025).

The most commonly used DL model in hybrids is LSTM, as many studies have shown, especially when combined with CNNs, attention mechanisms, transformers, other versions of neural networks (e.g., RNN), and hydrodynamic simulation (Cui et al. 2021; Alizadeh et al. 2021; Li et al. 2024; Oddo et al. 2024; An and Ouyang 2025). Additionally, ensemble architectures are commonly used in flood prediction contexts, yielding reliable and accurate predictions (Berkhahn et al. 2019) and deals with complex input interactions, complicated data, and has a higher learning capability compared to fundamental models (Kordani et al. 2024).

3.3 c. Input data

Choosing the right input is crucial for any ML/DL model, especially in a hydrological context, where the data can be variable and diverse (Zhou et al. 2023). Figure 8 shows the types of input data used in the included papers over the years.

Rainfall data is the most commonly used, followed by streamflow/discharge data and water level data; other types of data, such as temperature, hydrological data, DEM, and satellite data, are less commonly used. Each paper can include one or more datasets, as researchers often use multiple types of data for their task to achieve efficiency and accuracy in prediction. For example, Dhankhar et al. (2025a) used radar-based precipitation data to predict water levels. Xiang et al. (2025) used flow discharge data and rainfall data for flood probabilistic forecasting. Daramola et al. (2025) predicted extreme water levels using water level data from the European Centre for Medium-Range Weather Forecasts Reanalysis data set (ERA, version 5). An example of DEM employment in the flood prediction context is the work of Rana et al. (2024), which, due to limited datasets, employed high-resolution augmented DEMs generated with ConSinGAN. In a similar context, Situ et al. (2024) used spatial driving factors like DEM and drainage network along with rainfall series for urban flood predictions, while using geographic data and the Soil Conservation Service Curve Number (SCS-CN) method to predict runoff discharges.

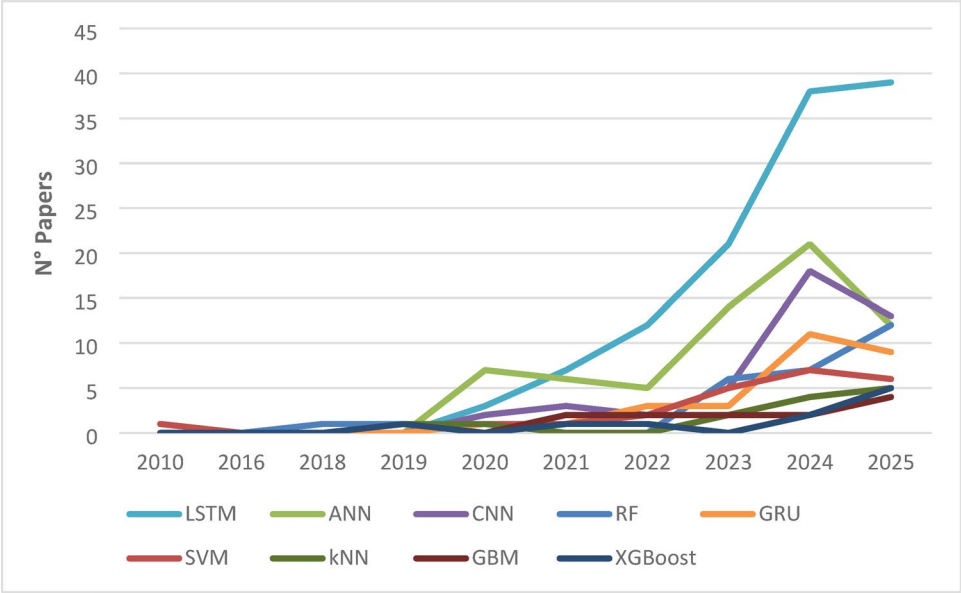


Fig. 7 Machine learning and Deep learning Models for flood prediction (2010–2025)

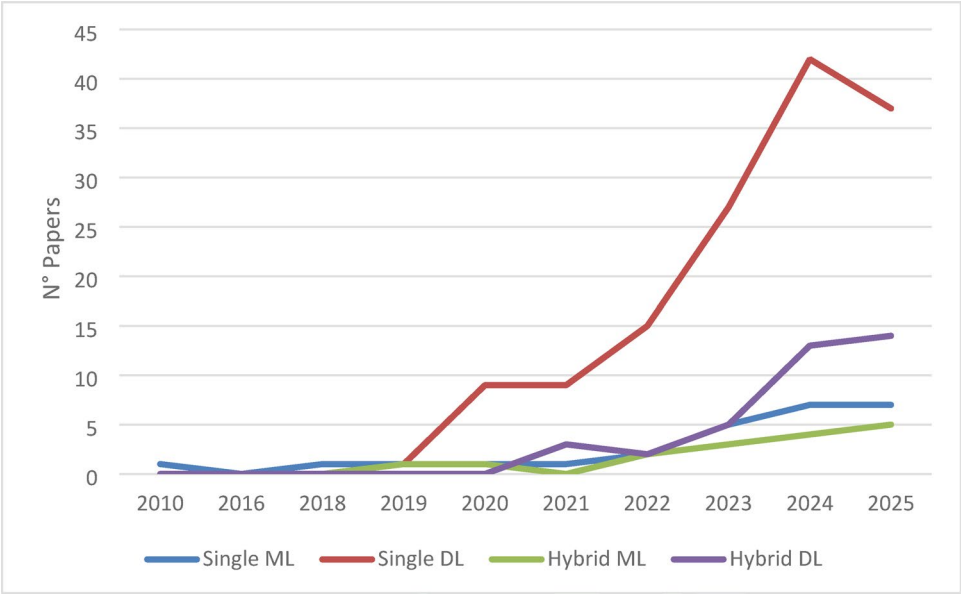


Fig. 8 Single and Hybrid ML/DL models (2010–2025)

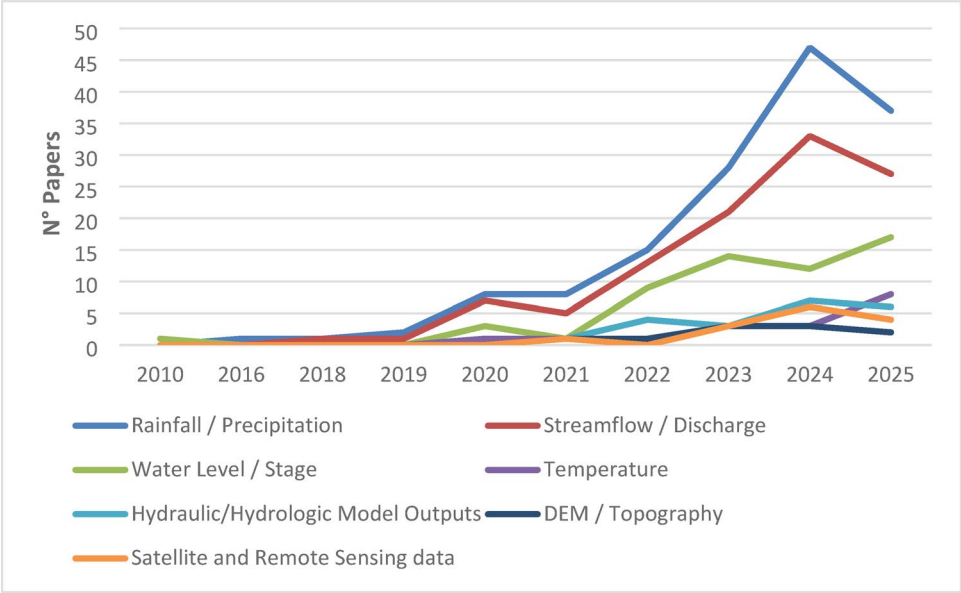


Fig. 9 Types of input data used for flood prediction (2010–2025)

3.4 d. Flood prediction tasks

Prediction tasks can be diverse, depending on the input data used, the methodology and models employed, and the context or geomorphology of the study area. Figure 9 illustrates how these prediction tasks disperse over the years. Three types of prediction tasks were identified: firstly, discharge and flow forecasting, which is the primary task among researchers when conducting a flood prediction study, considering flow discharge as the most vital hydrological indicator for water management and risk mitigation (Samantaray et al. 2023; Letessier et al. 2023). Secondly, peak time prediction is supported by a significant number of studies that have focused on it as their primary objective, which is also crucial for flood risk mitigation. It is often used in large-scale catchments and represents the time lag of streamflow runoff between the upstream and downstream segments (Chang and Tsai 2016; Ma et al. 2024). Water level prediction is often overlooked as a prediction task due to the rapid fluctuations of water levels, especially in small watersheds, which make it unpredictable and can produce noise that affects prediction accuracy. To address this issue, it is imperative to consider hybrid models along with sufficient meteorological and hydrological data (Chenmin et al. 2024; Dhankhar et al. 2025a; Wang et al. 2025).

LSTM, as the most widely used model in the studies, adopted discharge/flow forecasting as the primary prediction task, as Fig. 10 reveals, followed by nearly equal

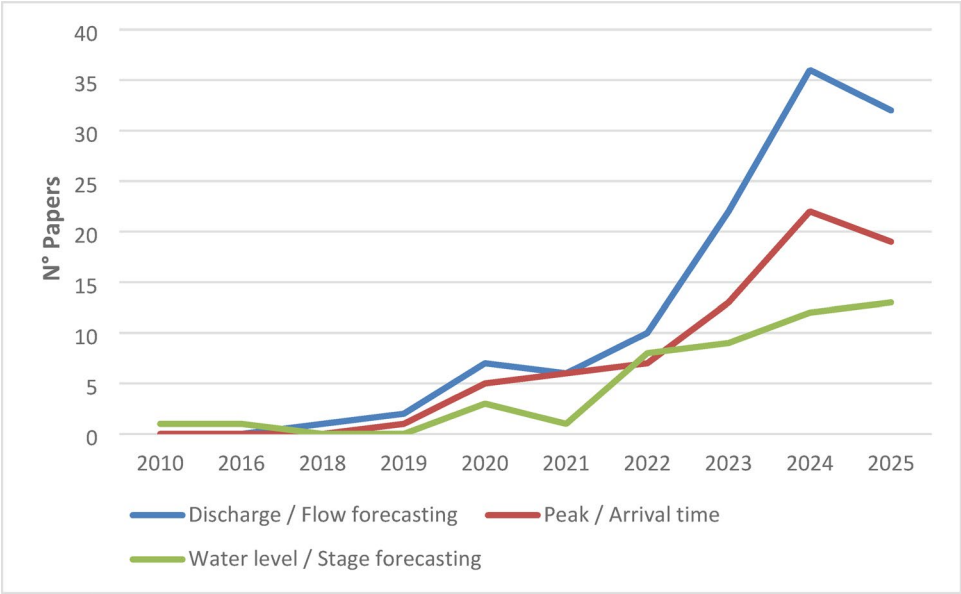


Fig. 10 Flood prediction tasks (2010–2025)

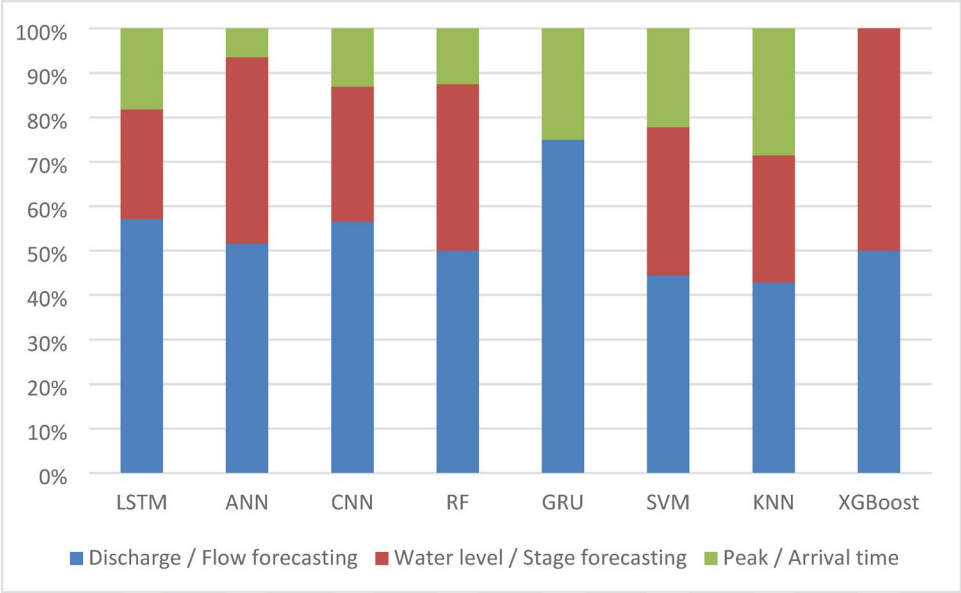


Fig. 11 ML/DL models and Flood prediction tasks

importance for water level and peak time forecasts. ANN has a balanced proportion between discharge and water level forecasting. In contrast, CNN-based model studies share a similar interest with LSTM-based model studies, but with a lesser focus on peak time forecasting. ML models, such as SVM, RF, and KNN, have a balanced usage across the three tasks, with RF receiving less importance attributed to peak time forecasting, particularly in the RF-based paper. GRU papers are heavily focused on discharge prediction and XGBoost prediction tasks, with an equal division between discharge and peak time forecasting.

3.5 e. Lead time

In flood forecasting, lead time is one of the most essential aspects. It defines a crucial parameter that can make a significant difference in risk mitigation, as it provides adequate time for flood warnings, emergency responses, and reduces the severity and probability of flood damage (Khatibi et al. 2003).

Figure 11 illustrates the evolution of lead time research over the years. Hourly, daily, and real-time predictions are considered short-term predictions, while weekly, monthly, and yearly predictions are considered long-term predictions. We can see that short-term prediction is the most widely adopted among researchers due to its importance in reducing

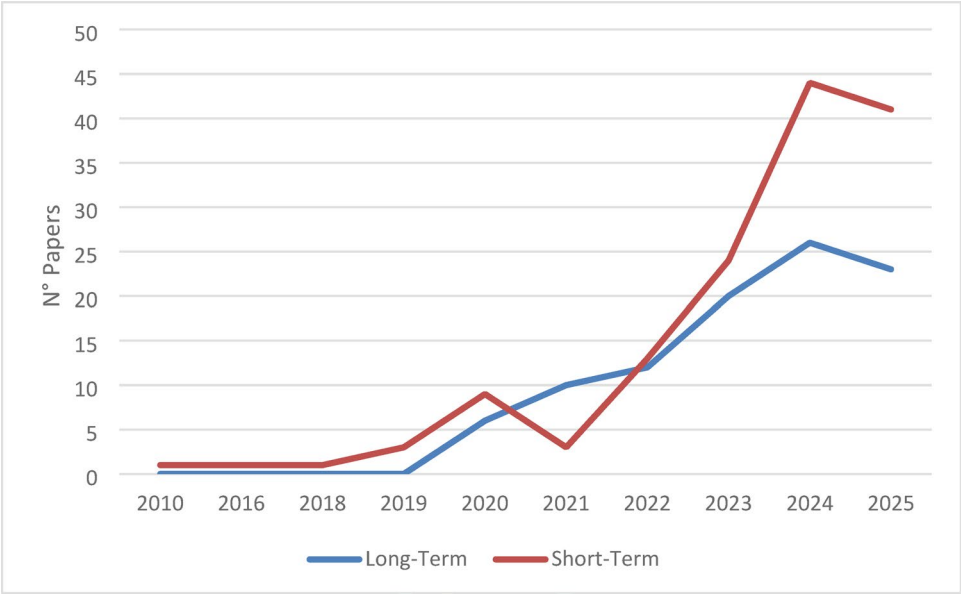
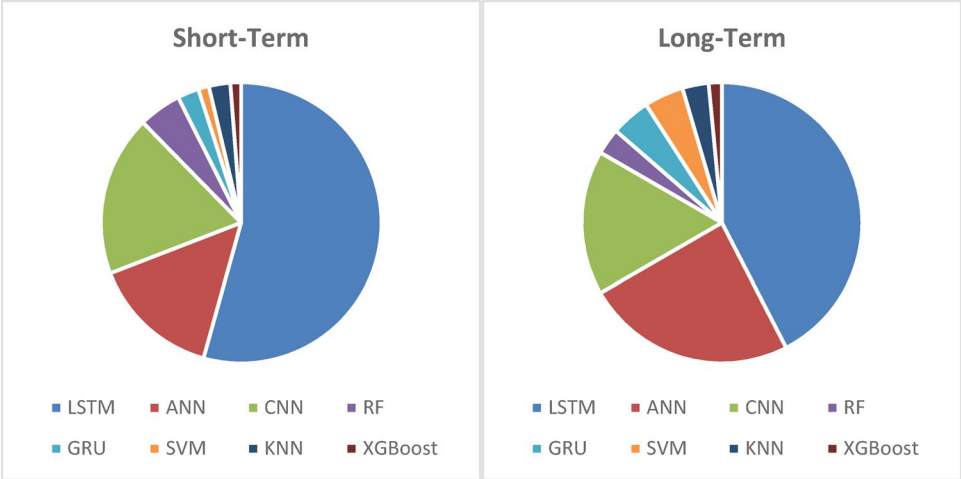


Fig. 12 Lead time for flood prediction (2010–2022)



(a) (b)

Fig. 13 ML/DL models and Lead time. **a** Short-Term. **b** Long-Term

damage, especially in urban areas and water resource management (Mosavi et al. 2018).

Long-term flood predictions are also significant in water resources management and can help decision-makers anticipate flood damage with infrastructure planning and reduce the impact of flooding (Sahoo et al. 2023; Hakim et al. 2024). However, it is a challenging task to conduct as it requires large-scale and sufficient time-series data that can increase noise, demanding precise and accurate methodologies (Cai and Yu 2022; Balthazar et al. 2024), which explains why its use among researchers is moderate compared to short-term lead forecasts. Figure 12 shows the distribution of ML/DL models for each lead time. It indicates that LSTM models dominate, followed by ANN and CNN models, while the other models share equal minimal proportions. In comparison between the two lead times, we can see that LSTM is more commonly used in short-term-based papers than in long-term-based ones, while SVM, GRU, and ANN models show an advantage in use in long-term lead papers compared to short-term lead papers.

4 Performance metrics

In this section, we will examine how performance metrics have been utilised to evaluate the models presented in the included papers that have evolved between 2010 and 2025. Figure 13 illustrates the most commonly reported performance metrics and their evolution over time. In contrast, Fig. 14 shows the relation between each LSTM and the performance metric in terms of usage percentage in the papers.

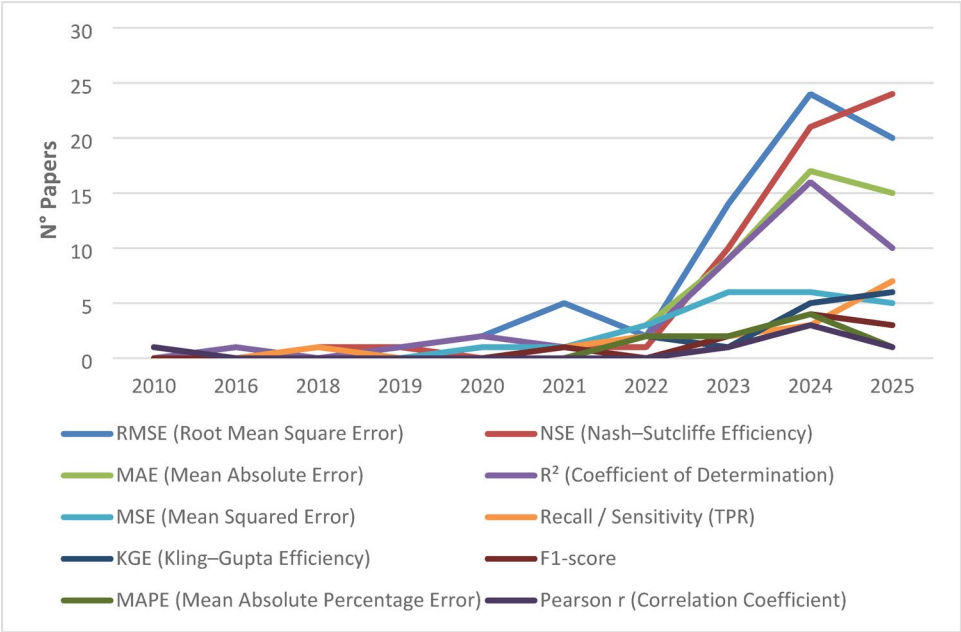


Fig. 14 Performance metrics for flood prediction (2010–2025)

Root Mean Square Error (RMSE) is the most reported metric in parallel with Nash–Sutcliffe Efficiency (NSE), followed by both Mean Absolute Error (MAE) and R^2. Other metrics, such as MSE, TPR, KGE, F1-score, MAPE, and Pearson's r coefficient, have been less commonly used to evaluate model performance over the years. In terms of use for each category of model, LSTM-based studies show an equal adoption of RMSE and NSE as primary metrics. In contrast, other metrics, such as MAE, KGE, and $R^{2,\ are}$ less frequently adopted, reflecting a preference for these metrics in time-series and streamflow forecasting studies. Similarly, ANN-based and CNN-based papers use the same proportion of metrics, with less adoption of the KGE metric as the main metric. The RMSE metric is strongly related to GRU-based papers, which are mainly used in conjunction with MAE for performance evaluation. In contrast, kNN-based papers rely solely on NSE for evaluation. R^2 and F1-score appear regularly in XGBoost, SVM, and RF-based papers due to their importance in evaluating regressions and flood probability studies (Fig. 15).

5 Research gap and future directions:

Our systematic review provides a current state of flood prediction studies using ML/DL models. It seeks to close an important research gap in the field and can address future recommendations and directions. Most research focuses on discharge/flow forecasting and water level as its primary tasks; however, given the importance of peak flow times and probabilistic flood occurrences in early warning and evacuation procedures, this area remains underexplored in research. Most studies focus on short-term lead predictions, leaving long-term forecasts underrepresented; this can affect the possibility of developing effective long-term water resources management strategies and limit their applications. Similar input data are used in the majority of studies, based on rainfall, gauge-based, and streamflow data, leaving a gap in exploring other possible and advanced data sources, such as remote sensing, high-resolution DEMs, and global climate indices (NAO, ENSO).

Regarding our most important factor, which is the ML/DL models used across the field, LSTM-based models dominate, leaving the opportunity for future research to explore different hybrid approaches and frameworks. Transformers, physically-based hydrodynamic modelling embedded with promising DL models and advanced neural network architectures, are highly recommended for the flood prediction task. In addition, not constraining performance evaluation to the most common metrics, such as RMSE and NSE, is highly encouraged. Using diverse metrics, such as KGE, can provide more clarity regarding the uncertainty and reliability of the prediction. To ensure transparency with readers, this study acknowledges its own limitations. Our review is limited to the Scopus Database, as it is the world's largest database of academic literature. Considering the rapid development of the field, the results of the research query adapted in the study, in terms of the number of papers, may be higher, which could affect the conclusions and results when replicating the same methodology. Our study is a systematic review. The classification of models, prediction tasks, input data, and lead times is purely statistical and systematic, based on use frequency and cross-evaluations between models and other key factors.

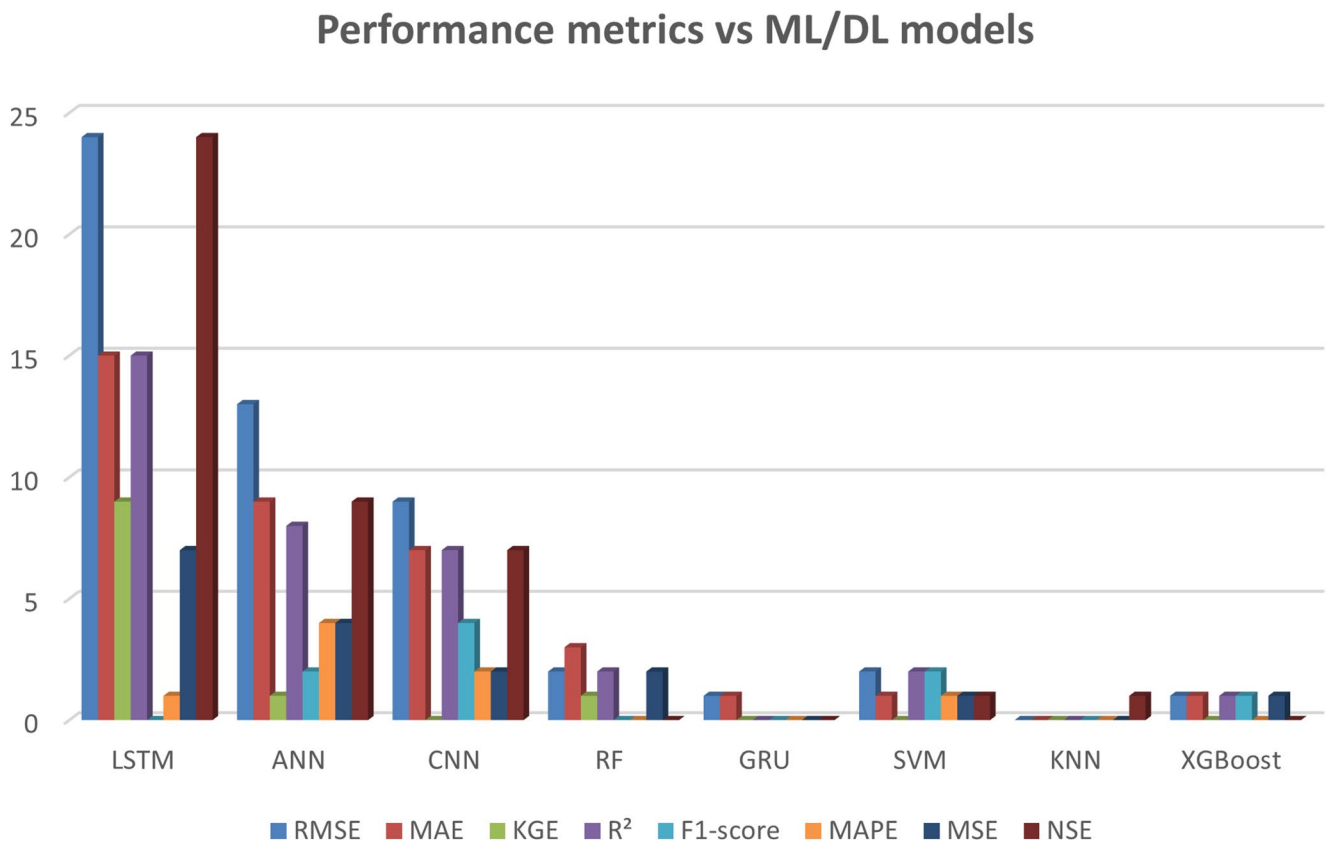


Fig. 15 ML/DL models and Performance metrics

6 Conclusions

This study presents an updated state of flood prediction field using ML and DL models. It gives insight into the most commonly used models and approaches, explores different prediction tasks, input data, and performance metrics. The study is conducted to synthesise papers published between 2005 and 2025 using a predefined methodology based on the PRISMA statement's four steps. The investigation was conducted on a total of 2698 papers in primary research, which were manually narrowed down to 237 papers through an eligibility evaluation based on predefined inclusion/exclusion criteria. The study has demonstrated the field's evolution over the years, with a notable increase in published papers over the last 4 years (2021–2024). Demonstrating a greater interest in finding solutions and developing new, precise, and accurate methods will reduce uncertainty and increase the reliability of flood predictions. The study revealed the dominance of LSTM-based models in the field, particularly for short-term predictions, while CNN and neural network-based models have gained increasing use in recent years. Hybrid and ensemble models have also shown significant interest among researchers, especially for complex inputs. The most commonly used predictors are rainfall and streamflow, which are employed in almost all the papers reviewed. The most frequently used metrics for performance evaluation are RMSE and NSE, with other metrics being gradually adopted over the years. This review contributed to a comprehensive overview of flood forecasting using ML and DL models, including all aspects that interfere with the methodologies and approaches adopted among researchers. It provides valuable insight into how the field originated, where it is headed, and can guide future research, ultimately improving the quality and accuracy of flood predictions to inform the right operational actions.

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Data availability The data that support the findings of this study are available on request from the corresponding author.

Declarations

Conflict of interests The authors declare no relevant financial or non-financial interests.

AI use statement The authors acknowledge the use of Large Language Models (LLMs) as an AI-assisted copy editing tool to improve human-generated texts for readability and style, ensuring they are free from errors in grammar, spelling, punctuation, and tone.

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